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Week 02: Monday, AST 5011: Astrophysical Systems

Virial Theorem, Constant-Density Model, Energy Generation/Transport, Dimensional Analysis, and the Main Sequence

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Self-Gravitating Objects

What Does "Self-Gravitating" Mean?

We often treat gravity as an externally imposed constant field $g = \text{const}$. This is *not* true for stars: a star's gravity is generated by its own mass distribution.

We will model a star as a self-gravitating fluid under the assumptions:

- No rotation
- Total mass

$$M_{\star} = \text{const}$$

(no mass loss)

The governing equations are conservation laws for:

- Mass
- Momentum
- Energy

These are the Euler equations.

Euler Equations (Fluid Description)

Conservation Law (Generic Form)

For a conserved quantity u with flux $\mathbf{F}$:

$$\partial_t u + \nabla \cdot \mathbf{F} = 0.$$

Euler Equations in 1D (Cartesian (x))

Mass (Continuity Equation)

$$\partial_t \rho + \partial_x(\rho v) = 0.$$

Momentum

$$\partial_t(\rho v) + \partial_x(\rho v^2 + P) = 0.$$

Energy

$$\partial_t \left(\frac{1}{2} \rho v^2 + \varepsilon \right) + \partial_x \left[v \left(\frac{1}{2} \rho v^2 + \varepsilon + P \right) \right] = 0.$$

Here:

- ε is the internal energy density
 - An equation of state (EOS) is required to relate P , ρ , and ε
-

Momentum Conservation with Gravity

Lagrangian (Co-moving) Form

In co-moving coordinates, the momentum equation becomes:

$$\rho \frac{d\vec{v}}{dt} = -\nabla P + \rho \vec{g}.$$

For a self-gravitating, spherically symmetric star:

$$\vec{g} = -\frac{Gm(r)}{r^2} \hat{r}.$$

Dynamical (Free-Fall) Timescale

Suppose pressure support were suddenly removed. The star would collapse under gravity.

In the radial direction:

$$\frac{d^2 r}{dt^2} = -\frac{GM}{R^2}.$$

A characteristic **free-fall timescale** is then:

$$t_{\text{ff}} \sim \sqrt{\frac{R^3}{GM}}.$$

Using the mean density ($\langle \rho \rangle = \frac{3M}{4\pi R^3}$):

$$t_{\text{ff}} \sim \frac{1}{\sqrt{G\langle \rho \rangle}}.$$

Key point: Stars evolve on timescales much longer than t_{ff} , so they must be extremely close to hydrostatic equilibrium.

The Virial Theorem

Motivation

The virial theorem provides a deep and surprisingly general connection between dynamics and energetics in self-gravitating systems. For stars, it allows us to relate gravity, pressure support, internal energy, and stability *without* knowing the detailed internal structure.

This makes it one of the most powerful tools in stellar astrophysics.

Hydrostatic Equilibrium

The virial theorem is the *integrated consequence* of hydrostatic equilibrium. It relates gravity to internal (thermal or radiation) energy.

$$\frac{dP}{dr} = -\rho \frac{Gm(r)}{r^2}, \quad \frac{dm}{dr} = 4\pi r^2 \rho(r).$$

Statement of the Virial Theorem

Following the discussion in *Hansen, Kawaler & Trimble (HKT) §1.3*, we arrive at the form:

$$\frac{1}{2} \frac{d^2 I}{dt^2} = 2K + \Omega$$

where:

- $I \equiv \sum_i m_i r_i^2$ is the moment of inertia of the system
- K is the total kinetic energy
- Ω is the gravitational potential energy

All quantities are summed over the entire star.

Physical Interpretation

- The left-hand side measures how the size of the system is changing in time.
- The right-hand side compares motion (kinetic energy) against gravity.

This equation is exact for any bound system of particles interacting via gravity.

Virial Equilibrium

For a star in a steady, long-lived state:

$$\frac{d^2 I}{dt^2} = 0$$

This defines virial equilibrium, giving:

$$2K + \Omega = 0$$

or equivalently:

$$K = -\frac{1}{2}\Omega$$

Key Consequences

- The total energy is

$$E = K + \Omega = \frac{1}{2}\Omega = -K$$

- Bound gravitational systems have negative total energy.
 - If gravity becomes stronger (more negative Ω), the system must increase its kinetic energy.
-

Connection to Temperature

In a star, most kinetic energy is thermal:

$$K \sim \frac{3}{2}NkT$$

Thus, via the virial theorem:

- Compression heats the star

- Expansion cools the star

This is the origin of a star's negative heat capacity.

In-class question: What happens to the temperature of a star if it loses energy by radiation?

► Answer

Order-of-Magnitude Estimate

Using:

$$\Omega \sim -\frac{GM^2}{R}$$

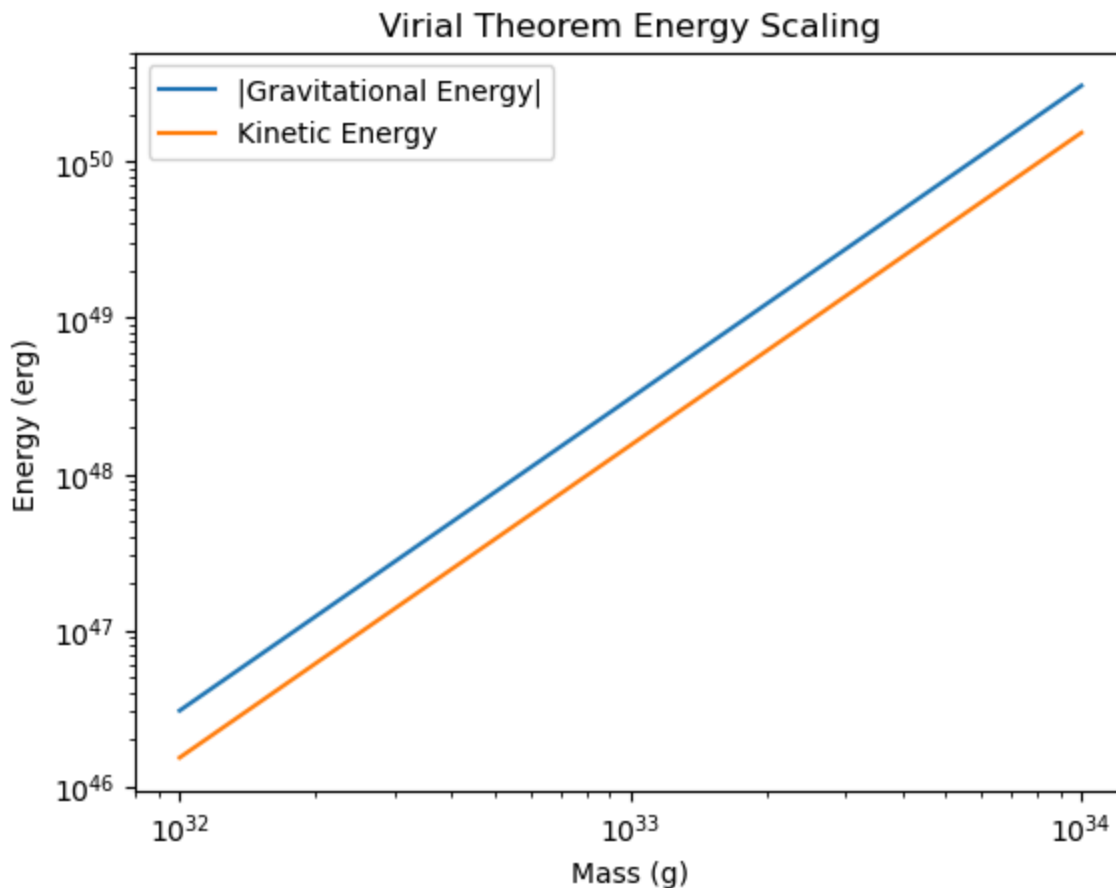
we find:

$$K \sim \frac{GM^2}{2R}$$

This estimate alone already tells us:

- Typical interior temperatures
- Why nuclear fusion is required
- Why stars contract as they cool

All without solving the full stellar structure equations.



In-class question: Why does changing the stellar radius shift both curves together?

► Answer

In-class Exercise: Virial Theorem and Stellar Heating

In this problem, you will explore how gravitational binding energy and thermal energy scale with stellar mass and radius, and connect this to the virial theorem and negative heat capacity.

Recall:

$$\Omega \sim -\frac{GM^2}{R}, \quad K = -\frac{1}{2}\Omega$$

Part A: Energy Scaling

- Using the expressions above, compute the gravitational potential energy Ω and kinetic energy K for a range of stellar masses.
- Assume a fixed stellar radius and plot:
 - $|\Omega|$ vs mass
 - K vs mass
- Verify that both energies scale as M^2 .

Part B: Effect of Contraction

Now fix the stellar mass and vary the radius.

1. Plot $|\Omega|$ and K as a function of radius.
2. Describe what happens to the kinetic energy as the star contracts.
3. Explain how this relates to the statement: "A star heats up as it loses energy."

Part C: Conceptual Question

Using your plots and the virial theorem, answer:

Why does a self-gravitating system have negative heat capacity?

► Solution

Limits of Applicability

The simple virial form assumes:

- The system is bound
- Surface terms are negligible
- No rapid mass loss or violent dynamics

It does *not* apply directly to:

- Supernova explosions
- Strongly driven stellar winds
- Rapidly accreting or eruptive systems

Why the Virial Theorem Matters

The virial theorem:

- Explains why stars shine by gravitational contraction (Kelvin–Helmholtz phase)
- Predicts negative heat capacity
- Sets the scale for stellar temperatures
- Applies equally to stars, star clusters, galaxies, and dark-matter halos

It is one of the few tools in astrophysics that is both simple and profound.

Looking Ahead

Next, we will use the virial theorem to understand:

- Kelvin–Helmholtz contraction timescales
- The onset of nuclear burning
- Stability of self-gravitating objects

This will connect dynamics directly to stellar evolution.

Equations of Stellar Structure

Motivation

We are now close to being able to construct simple models of stars. To do this, we need a complete set of equations that relate:

- Mass and radius
- Pressure and gravity
- Energy generation and transport
- Thermodynamic state of the gas

Once assembled, these equations allow us (in principle) to solve for the internal structure of a star. Over the next several lectures, we will focus on the individual physical ingredients: energy generation, equations of state, and opacities.

Energy Transport: Physical Picture

We begin with energy transfer. What we care about is the net flux of energy at a given location inside the star.

Because we assume spherical symmetry, the flux can only have a radial component.

In stellar interiors, particle mean free paths are typically very small compared to the stellar radius. Energy is therefore transported by diffusion: particles scatter many times before escaping.

In-class question: If the energy density were constant everywhere in a star, would there be any net energy flux?

► Answer

This immediately tells us that flux must depend on the gradient of the energy density:

$$F \propto \frac{dE}{dr}$$

In-class question: Should the flux increase or decrease if the particle mean free path ℓ increases?

► Answer

Because opacity (κ) is inversely related to mean free path, this suggests:

$$F \propto \kappa^{-1} \frac{dE}{dr}$$

In-class question: If outward flux is positive, what sign must appear in front of the gradient?

► Answer

Radiative Energy Transport

We now specialize to radiation as the energy-carrying agent. Other mechanisms exist (convection, conduction), but radiation dominates large fractions of many stars.

The radiation energy density is:

$$E = aT^4$$

Including the effects of diffusion, the radiative flux is:

$$F_r = -\frac{c}{3\kappa\rho} \frac{d(aT^4)}{dr}$$

Multiplying by the surface area of a sphere gives the luminosity:

$$L_r = -\frac{4\pi r^2 c}{3\kappa\rho} \frac{d(aT^4)}{dr}$$

This is the radiative energy transport equation.

Physical Meaning of the Terms

- κ has units of

$$\text{cm}^2 \text{g}^{-1}$$

and represents the opacity.

- $\frac{1}{\kappa\rho}$ has units of length and is the mean free path.
- The factor of

$$c$$

appears because radiation travels at the speed of light.

In-class question: Why does faster particle speed imply larger energy flux, all else equal?

► Answer

Lagrangian Form

Using:

$$dM_r = 4\pi r^2 \rho dr$$

we can rewrite the transport equation as:

$$L_r = -\frac{(4\pi r^2)^2 c}{3\kappa} \frac{d(aT^4)}{dM_r}$$

This form is often more convenient for stellar structure calculations.

Opacity: Power-Law Approximation

Although real opacities are complicated, they are often approximated as:

$$\kappa = \kappa_0 \rho^n T^{-s}$$

Examples:

- **Thomson scattering:**

$$n = 0, s = 0$$

- **Kramers opacity:**

$$n = 1, s = 3.5$$

These approximations are valid only over limited ranges of density and temperature.

Equation of State

We also require a relation between pressure, density, and temperature:

$$P = P_0 \rho^{\chi_\rho} T^{\chi_T}$$

For an ideal gas:

$$\chi_\rho = 1, \quad \chi_T = 1$$

Similarly, energy generation is often written as:

$$\epsilon = \epsilon_0 \rho^\lambda T^\nu$$

All of these are approximations, but they are extremely useful.

The Four Equations of Stellar Structure

Together, we arrive at four fundamental equations:

Mass Conservation

$$\frac{dM_r}{dr} = 4\pi r^2 \rho$$

Hydrostatic Equilibrium

$$\frac{dP}{dr} = -\frac{GM_r \rho}{r^2}$$

Energy Generation

$$\frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon$$

Energy Transport (Radiative)

$$L_r = -\frac{4\pi r^2 c}{3\kappa \rho} \frac{d(aT^4)}{dr}$$

These four equations, combined with boundary conditions, define a stellar model.

In-class Exercise: Radiative Energy Transport and Temperature Gradients

In this exercise, you will explore how radiative energy transport sets the temperature gradient inside a star.

Starting from the radiative transport equation:

$$L_r = -\frac{4\pi r^2 c}{3\kappa \rho} \frac{d(aT^4)}{dr}.$$

Part A: Rearranging the Transport Equation

1. Starting from the equation above, derive an expression for dT/dr in terms of L_r , r , κ , ρ , and T .
 2. Identify which quantities act to steepen the temperature gradient.
-

Part B: Scaling with Radius

Assume:

- constant opacity κ ,
- constant density ρ ,
- luminosity increases outward as $L_r \propto r^3$.

1. Using your expression for dT/dr , determine how the temperature gradient scales with radius.
 2. Explain why the gradient must be steep near the stellar center.
-

Part C: Numerical Exploration

Using the code below:

1. Compute the luminosity profile $L_r(r)$ for a given temperature profile $T(r)$.
2. Identify where $|dT/dr|$ is largest.
3. Verify that this corresponds to the smallest radii.

► Solution

Dimensional Analysis and Homology Relations

Philosophy

Homology relations assume that stars are scaled versions of a single reference star, with:

- Identical composition
- Identical microphysics
- No preferred physical scale

This implies that many quantities must be **power laws**.

Key idea: Scale-free systems imply power-law relations.

Homologous Scaling

If a reference star has mass

$$M_0$$

and radius

$$R_0$$

, we assume:

$$r = \frac{R}{R_0} r_0, \quad M_r = \frac{M}{M_0} M_{r,0}$$

Assuming:

$$R \propto M^{\alpha_R}, \quad \rho \propto M^{\alpha_\rho}$$

The mass equation implies:

$$\alpha_\rho + 3\alpha_R = 1$$

Example: Main Sequence Scaling

For moderately massive main-sequence stars:

- Electron scattering opacity
- Ideal gas equation of state
- CNO-cycle hydrogen burning

Homology predicts:

$$\frac{R}{R_\odot} \propto \left(\frac{M}{M_\odot} \right)^{0.78}, \quad \frac{L}{L_\odot} \propto \left(\frac{M}{M_\odot} \right)^3$$

Empirically, the luminosity exponent is closer to

$$3.5$$

— remarkably good agreement given the assumptions.

Main-Sequence Lifetime Estimate

Assuming a fixed fraction of mass is available for fusion:

$$\tau_{\text{MS}} \propto \frac{M}{L}$$

Using the empirical relation:

$$\tau_{\text{MS}} \approx 10^{10} \left(\frac{M}{M_{\odot}} \right)^{-2.5} \text{ yr}$$

In-class question: Why do massive stars live such short lives despite having more fuel?

► Answer

Limitations of Homology

Homology relations fail when:

- Composition varies
- Degeneracy pressure dominates
- Radiation pressure becomes important
- Opacities change rapidly

In-class question: Which assumptions above break first for very low-mass stars? Very high-mass stars?

► Answer

In-class Exercise: Dimensional Analysis and Homology Scaling

In this exercise, you will use dimensional analysis and homology arguments to derive basic stellar scaling relations and explore their limitations.

Recall that homology assumes:

- identical composition and microphysics,
 - no preferred physical scale,
 - stars differ only by an overall rescaling of mass and radius.
-

Part A: Density and Radius Scaling

Assume the stellar radius and density scale with mass as:

$$R \propto M^{\alpha_R}, \quad \rho \propto M^{\alpha_\rho}.$$

1. Starting from mass conservation,

$$M \sim \rho R^3,$$

derive a constraint relating α_R and α_ρ .

2. Explain why only one of these exponents is independent.

Part B: Luminosity Scaling

Assume:

- energy generation per unit mass scales as

$$\epsilon \propto \rho^\lambda T^\nu,$$

- luminosity scales as

$$L \sim M\epsilon.$$

1. Using dimensional arguments, explain why the luminosity must be a power law in mass.
2. If $L \propto M^3$, estimate how the luminosity changes when the mass doubles.

Part C: Main-Sequence Lifetime

Assume a fixed fraction of the stellar mass is available for nuclear burning.

1. Show that the main-sequence lifetime scales as

$$\tau_{\text{MS}} \propto \frac{M}{L}.$$

2. Using $L \propto M^{3.5}$, determine how the lifetime changes for a star with $2M_\odot$ compared to the Sun.

Part D: Breakdown of Homology

For each case below, identify which homology assumption fails and why:

1. Very low-mass stars
2. Very high-mass stars

Relate your answer to changes in pressure support, opacity, or energy transport.

► Solution

The Constant-Density Model

For the following analysis we will make the assumption that the central density is the density throughout the entire star, $\rho = \rho_c = \text{constant}$ and the c subscript denotes the center.

In HKT 1.4, we saw that making this approximation allows us to, with some "trivial" algebra come up with an equation for the central pressure (*under the above assumption*)

$$P_c = \frac{3}{8\pi} \frac{GM_\star^2}{R_\star^2} = 1.34 \times 10^{15} \left(\frac{M_\star}{M_\odot} \right)^2 \left(\frac{R_\star}{R_\odot} \right)^{-4} (\text{dyne cm}^{-2}).$$

Calculation of Molecular Weights

Some definitions and vocabulary:

- n - the number of free particles in a gas
- Z - nuclear charge
- A - nuclear mass number in (amu)
- X - fraction by mass of a species (aka mass fraction)
- $n_{I,i}$ - the ion number density in units of cm^{-3} of a given species i

The ion number density takes the form of (HKT 1.40)

$$n_{I,i} = \frac{(\text{mass/unit volume of species } i)}{(\text{mass of 1 ion of species } i)} \equiv \frac{\rho X_i N_A}{A_i}$$

Taking a sum over all ions gives the and defining μ_I as the total mean molecular weight of ions, we have the ion number density for all ions as

the ion number density for all ions

$$n_I = \frac{\rho N_A}{\mu_I}$$

Here, $N_A = 6.022137 \times 10^{23} \text{ mole}^{-1}$.

As such, we can rewrite

mean molecular weight of ions

$$\mu_I = \left[\sum_i \frac{X_i}{A_i} \right]^{-1}$$

Following a similar derivation, we can also define the mean molecular weight per free electron,

mean molecular weight per free electron

$$\mu_e = \left[\sum_i \frac{Z_i X_i y_i}{A_i} \right]^{-1}$$

where y_i is the ionization fraction, a value of $y_i = 1$ means the gas is completely ionized while $y_i = 0$ implies the gas is completely neutral.

We finally arrive at expressions for the total mean molecular weight,

total mean molecular weight

$$\mu = \left[\frac{1}{\mu_I} + \frac{1}{\mu_e} \right]^{-1}$$

with $n = n_I + n_e = \frac{\rho N_A}{\mu}$

Some Naming Conventions and Approximations

The mass fractions X_i for often defined as X for ^1H and Y for ^4He and all else referred to as "metals" denoted by Z (not ion charge!).

This leads to the following result for the composition of a star, $X + Y + Z = 1$.

Now, lets consider some scenarios where we can approximate values for the molecular weights.

When X, Y , and Z are completely ionized $y_i = 1$ and $Z \ll 1$ (metals are a small mass fraction)

$$\mu_e \approx \frac{2}{1 + X}$$

When using the same assumptions as above and the fact that Z is small compared the $\langle A \rangle$

$$\mu_I \approx \frac{4}{1 + 3X}$$

Combining these results leads to an approximate total mean molecular weight

$$\mu \approx \frac{4}{3 + 5X}$$

Energy Generation and Transport

Some definitions:

- ϵ - energy generation rate ($\text{erg g}^{-1} \text{ s}^{-1}$)
- κ - lets you know how the flow of radiation is hindered by the medium through which it passes ($\text{cm}^{-2} \text{ g}^{-1}$)

The energy equation

$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon$$

or in Lagrangian form:

$$\frac{dL}{dm} = \epsilon$$

Stellar Dimensional Analysis

In this section, we found the Mass-Luminosity relationship for stars with masses greater than about one solar mass,

$$\frac{R}{R_{\odot}} \approx \left(\frac{M}{M_{\odot}} \right)^{0.75}$$

$$\frac{L}{L_{\odot}} \approx \left(\frac{M}{M_{\odot}} \right)^{3.5}$$

Evolutionary Lifetimes on the Main Sequence

$$t_{\text{nuc}} \approx 10^{10} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L}{L_{\odot}} \right)^{-1} \text{ (years)}$$

or using the mass-luminosity relationship,

$$t_{\text{nuc}} \approx 10^{10} \left(\frac{M}{M_{\odot}} \right)^{-2.5} \text{ (years)}$$

In-class Exercise: Stellar Molecular Weights

Part A: Single Calculation (Manual)

Using the equations:

$$\mu_I = \left[\sum_i \frac{X_i}{A_i} \right]^{-1}, \quad \mu_e = \left[\sum_i \frac{Z_i X_i y_i}{A_i} \right]^{-1}, \quad \mu = \left[\frac{1}{\mu_I} + \frac{1}{\mu_e} \right]^{-1},$$

compute μ_I , μ_e , and μ for a fully ionized gas with the following composition:

Species	Mass fraction	A	Z
H	X = 0.70	1	1
He	Y = 0.28	4	2
Metals	Z = 0.02	12	6

Assume complete ionization ($y_i = 1$).

Questions:

1. Compute μ_I using the sum over X_i/A_i .
2. Compute μ_e using the sum over $Z_i X_i y_i / A_i$.
3. Compute the total mean molecular weight μ .
4. Compare your values to approximate formulas for a Sun-like composition:

$$\mu_I \approx \frac{4}{1 + 3X}, \quad \mu_e \approx \frac{2}{1 + X}, \quad \mu \approx \frac{4}{3 + 5X}.$$

Part B: Python Calculation

Write a Python function that takes arbitrary X, Y, Z values and computes:

- μ_I — mean molecular weight per ion
- μ_e — mean molecular weight per free electron
- μ — total mean molecular weight

Use the exact formulas:

$$\mu_I = \left[\frac{X}{1} + \frac{Y}{4} + \frac{Z}{\langle A \rangle_{\text{metals}}} \right]^{-1}$$

$$\mu_e = \left[\frac{X}{1} + \frac{Y}{2} + \frac{Z \langle Z \rangle_{\text{metals}}}{\langle A \rangle_{\text{metals}}} \right]^{-1}$$

$$\mu = \left[\frac{1}{\mu_I} + \frac{1}{\mu_e} \right]^{-1}$$

Part C: Conceptual Questions

- How does increasing helium fraction (smaller X) affect the total mean molecular weight?
- How would a higher μ influence the central temperature and fusion rate in a star?
- Explain why massive stars live much shorter lives than low-mass stars despite having more fuel.

► Solution

In-Class Assignment

Learning Objectives

- building stellar models using approximations

We will be picking up from the exercise from last class. These exercises require the solutions and are included here.

IPython console for SymPy 1.13.3 (Python 3.11.8-64-bit) (ground types: python)

These commands were executed:

```
>>> from sympy import *
>>> x, y, z, t = symbols('x y z t')
>>> k, m, n = symbols('k m n', integer=True)
>>> f, g, h = symbols('f g h', cls=Function)
>>> init_printing()
```

Documentation can be found at <https://docs.sympy.org/1.13.3/>

stellar model

Imagine a star has a density profile of the form:

$$\rho(r) = \rho_c \left[1 - \left(\frac{r}{R_\star} \right)^3 \right]$$

where ρ_c is the central density.

$$\rho_c \left(1 - \frac{r^3}{R_\star^3} \right)$$

We found expression for the central density in terms of $R_\star$ and $M_\star$.

$$\frac{2\pi R_\star^3 \rho_c}{3}$$

$$\frac{3M_\star}{2\pi R_\star^3}$$

$$\frac{3M_\star \left(1 - \frac{r^3}{R_\star^3} \right)}{2\pi R_\star^3}$$

1. - Computing Central Pressure

Our goal is to find an equation for the pressure as a function of radius $P(r)$ using the above $\rho(r)$ we computed.

Recall our equation for hydrostatic equilibrium (HSE):

$$\frac{dP}{dr} = -\rho(r)g(r)$$

where

$$g(r) = -\frac{GM(r)}{r^2}$$

Also assume that the pressure vanishes at the surface $P(R_\star) \rightarrow 0$.

Your answer will be in the form of $P(r) = P_c \times (\text{polynomial in } r/R_\star)$.

1. Integrate our mass equation now from 0 to some radius r (as opposed to 0 to $R_\star$ when looking for ρ_c). This should give us $M(r)$.
2. Compute $P(r)$ in terms of $M_\star$ and $R_\star$ by integrating our HSE equation from 0 to $R_\star$.
3. Solve for P_c by setting $P(R_\star) = 0$.
4. Compute a numerical value for P_c by substituting in values for $M_\star$ and $R_\star$ and compare to the Sun's central pressure of about 2×10^{17} (dyne cm⁻²).
5. Plot the resulting pressure profile $P(r)$ by defining a new function $z = P(r)/P_c$, then substitute (r for a fractional radius of the total radius $R_\star$ as $\xi \times R_\star$). Run the cells to plot the result.

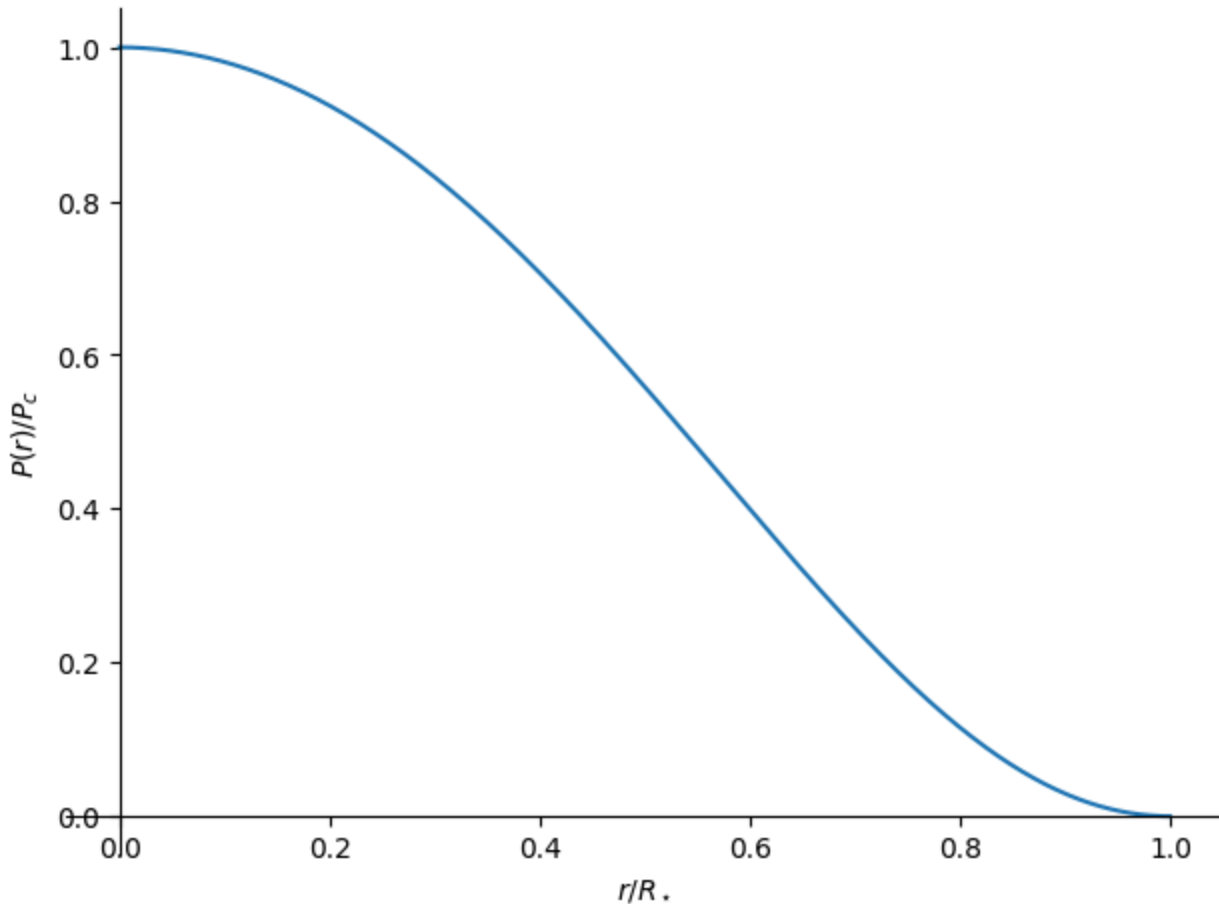
$$\frac{2M_\star r^3}{R_\star^3} - \frac{M_\star r^6}{R_\star^6}$$

$$-\frac{63GM_\star^2}{80\pi R_\star^4} + P_c$$

$$\frac{63GM_\star^2}{80\pi R_\star^4}$$

$$2.85003082072768 \cdot 10^{15}$$

$$\frac{63GM_\star^2}{80\pi R_\star^4} - \frac{3GM_\star^2 r^2}{2\pi R_\star^6} + \frac{9GM_\star^2 r^5}{10\pi R_\star^9} - \frac{3GM_\star^2 r^8}{16\pi R_\star^{12}}$$



<sympy.plotting.backends.matplotlibbackend.matplotlib.MatplotlibBackend at 0x12404b450>

2. - Computing Central Temperature

The central temperature, assuming the ideal gas law, is:

$$T_c = \frac{\mu m_u P_c}{\rho_c k}$$

For our density profile this should be

$$T_c = \frac{21}{40} \frac{GM_*}{R_*} \frac{\mu m_u}{k}$$

while for the constant density model, the coefficient was 1/2.

1. Compute T_c symbolically and confirm the coefficient 21/40.
2. Compute a numerical value for the central temperature and compare to the central temperature of the Sun.
- 3.

Okay great, but we still need to know the composition before we can compute the temperature.

We'll look at two choices here:

$$\frac{21GM_{\star}m_u\mu}{40R_{\star}k}$$

i. pure H composition has $\mu = 1/2$, compute T_c for this composition.

0.6052083333333333

ii. pure He composition has $\mu = 4/3$

1.613888888888889

The central pressure for the constant-density model (HKT 1.39) was

$$P_c = \frac{3}{8\pi} \frac{GM_{\star}^2}{R_{\star}^4}$$

Here, with the cubic density profile, we find

$$P_c = \frac{63}{80\pi} \frac{GM_{\star}^2}{R_{\star}^4}$$

The pressure is greater in this model because more mass is concentrated toward the center of the star, increasing the force of gravity throughout, making the outer layers weigh more.

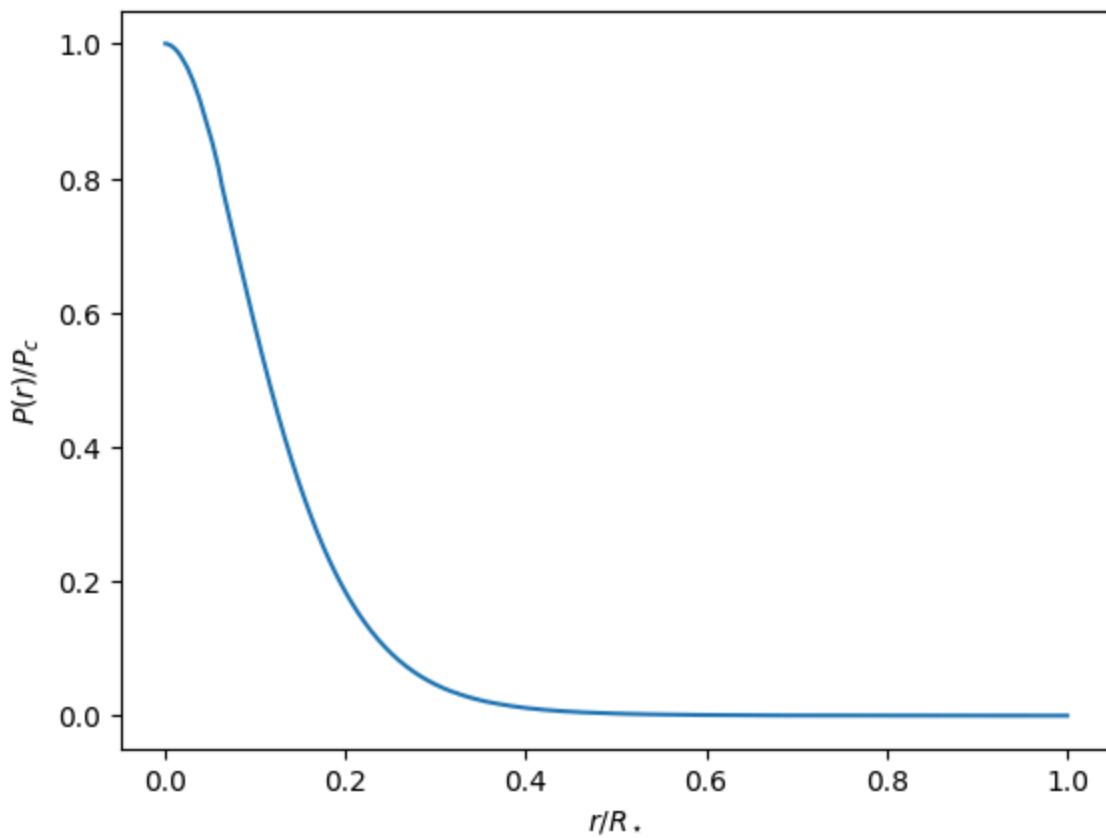
The temperature is the *almost the same* in both models, since both the central pressure and central density increase by the here.

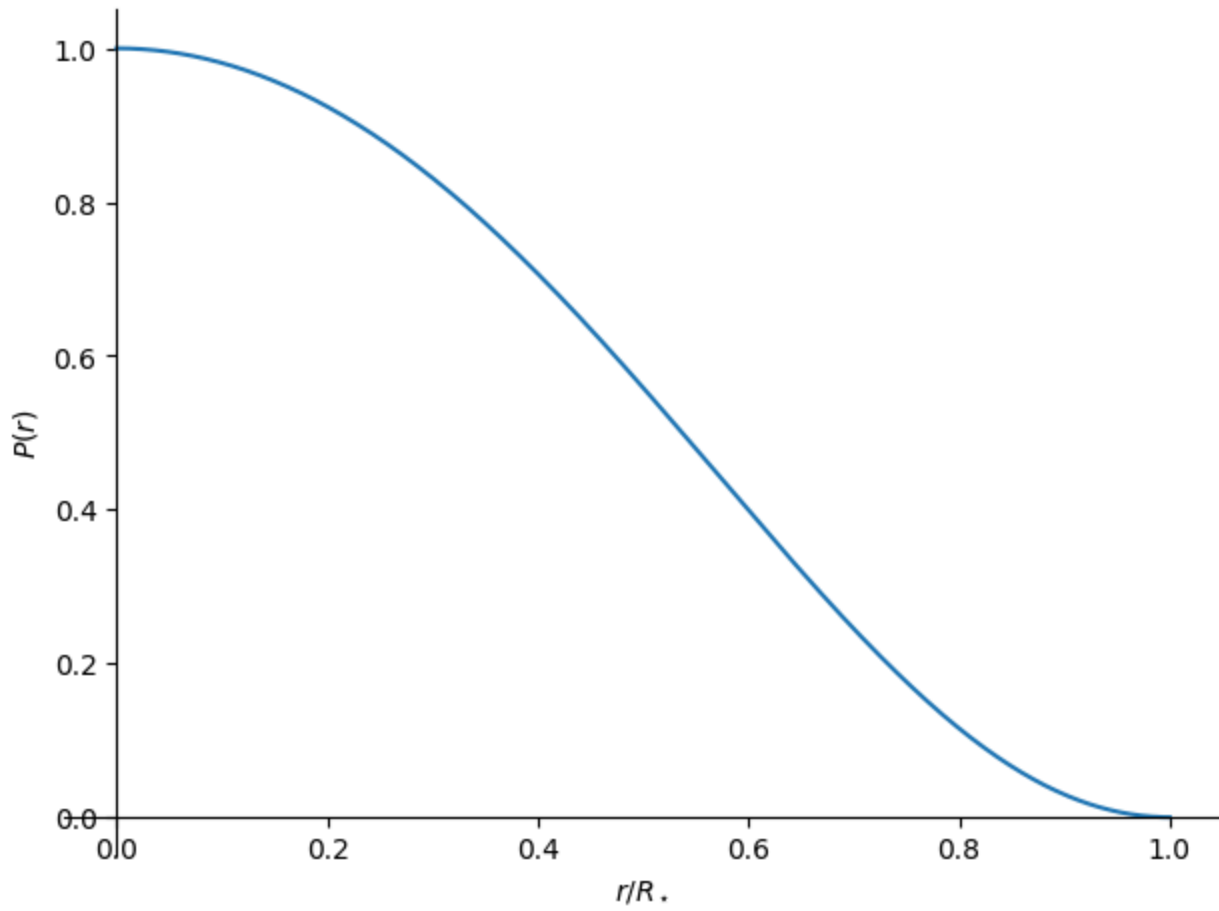
Let's evaluate P_c/ρ_c for our model

3. Comparing the Pressure Profiles to a MS stellar model

1. load the MESA data using pandas,
2. retrieve the radius and pressure profiles
3. compute the max / central pressure
4. plot the MESA data $r/R_{\star}$ vs $P(r)/P_c$ and compare directly to our plot for these data using our approximate density profile
5. comment briefly on the difference of the two profiles and think what might be contributing to the difference

```
Index(['mass', 'radius', 'luminosity', 'pressure', 'logRho', 'logT', 'energy',
      'entropy', 'cp', 'gamma1', 'grada', 'mu', 'free_e', 'ye', 'pgas',
      'prad', 'gradr', 'gradT', 'velocity', 'conv_vel', 'opacity', 'eps_nuc',
      'pp', 'cno', 'tri_alfa', 'eps_nuc_neu_total', 'non_nuc_neu', 'eps_grav',
      'h1', 'he3', 'he4', 'c12', 'n14', 'o16', 'ne20', 'mg24', 'si28', 's32',
      'ar36', 'ca40', 'ti44', 'cr48', 'fe52', 'fe54', 'fe56', 'ni56', 'eta',
      'log_omega', 'v_rot', 'j_rot', 'dynamo_log_B_r', 'dynamo_log_B_phi',
      'log_D_conv', 'log_D_semi', 'log_D_ovr', 'log_D_thrm'],
      dtype='object')
```





<sympy.plotting.backends.matplotlibbackend.matplotlib.MatplotlibBackend at 0x2a30f21d0>

comment on the difference for the two pressure profiles here...